

ELI — The Complete User Manual

Plain-English Edition · with flowcharts

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This manual assumes **zero** technical background. You talk to ELI in normal English — by **typing** in the chat box or **speaking** — and it does things on your computer, answers questions, remembers what matters to you, and learns your routines. Everything runs **on your own machine**; nothing is sent to the cloud.

The one rule worth knowing: ELI is **offline by default**. It only touches the internet when *you* ask for something that needs it (a web search, the news, downloading a model). Your conversations, files, and memories never leave your computer.

How to read this manual: - **Just want to use it?** Read §1–§5 and the cheat sheet (§18). That's enough to be productive. - **Want the whole picture?** Read straight through — every feature is here, in plain terms. - **[Anyone] / [A bit technical] / [Advanced] tags** at the start of some sections mean *anyone* / *a little technical* / *advanced*.

1. What ELI is (in one minute)

ELI is a **personal AI assistant that lives on your computer**. Think of it as a capable helper that can:

- **Talk with you** like a knowledgeable person — by text or voice.
- **Operate your computer** — open apps, play music, manage windows, type for you.
- **Look at things** — your screen, images, PDFs, spreadsheets — and explain them.
- **Write for you** — documents, notes, code, scripts.
- **Remember** what matters about you, and **learn** your daily routines.
- **Act on its own** when helpful — a morning briefing, a heads-up, a suggestion.

The big difference from cloud assistants: **it's yours**. It runs locally, stays private, has no subscription, and you can even retrain its “voice” yourself (\$14).

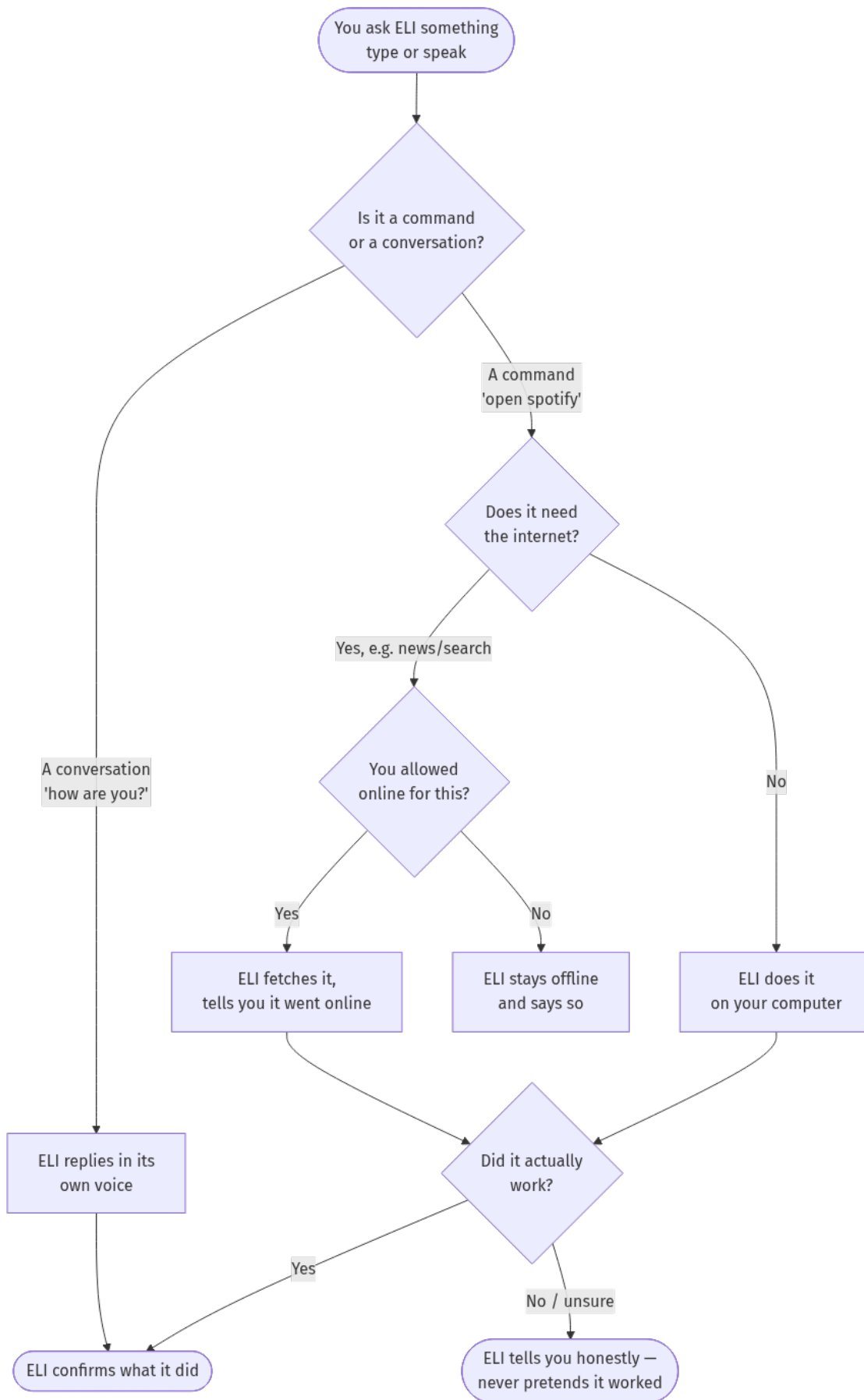
Here's the whole thing at a glance:



2. How ELI works when you ask it something

[Anyone] Anyone — *this is the mental model that makes everything else click.*

When you say or type something, ELI follows the same honest loop every time:



The promise in that diagram: ELI **never fakes an action**. If it says it played a song or fixed a file, it really did — and if it couldn't, it says so plainly instead of bluffing. This is built into the software, not just good manners.

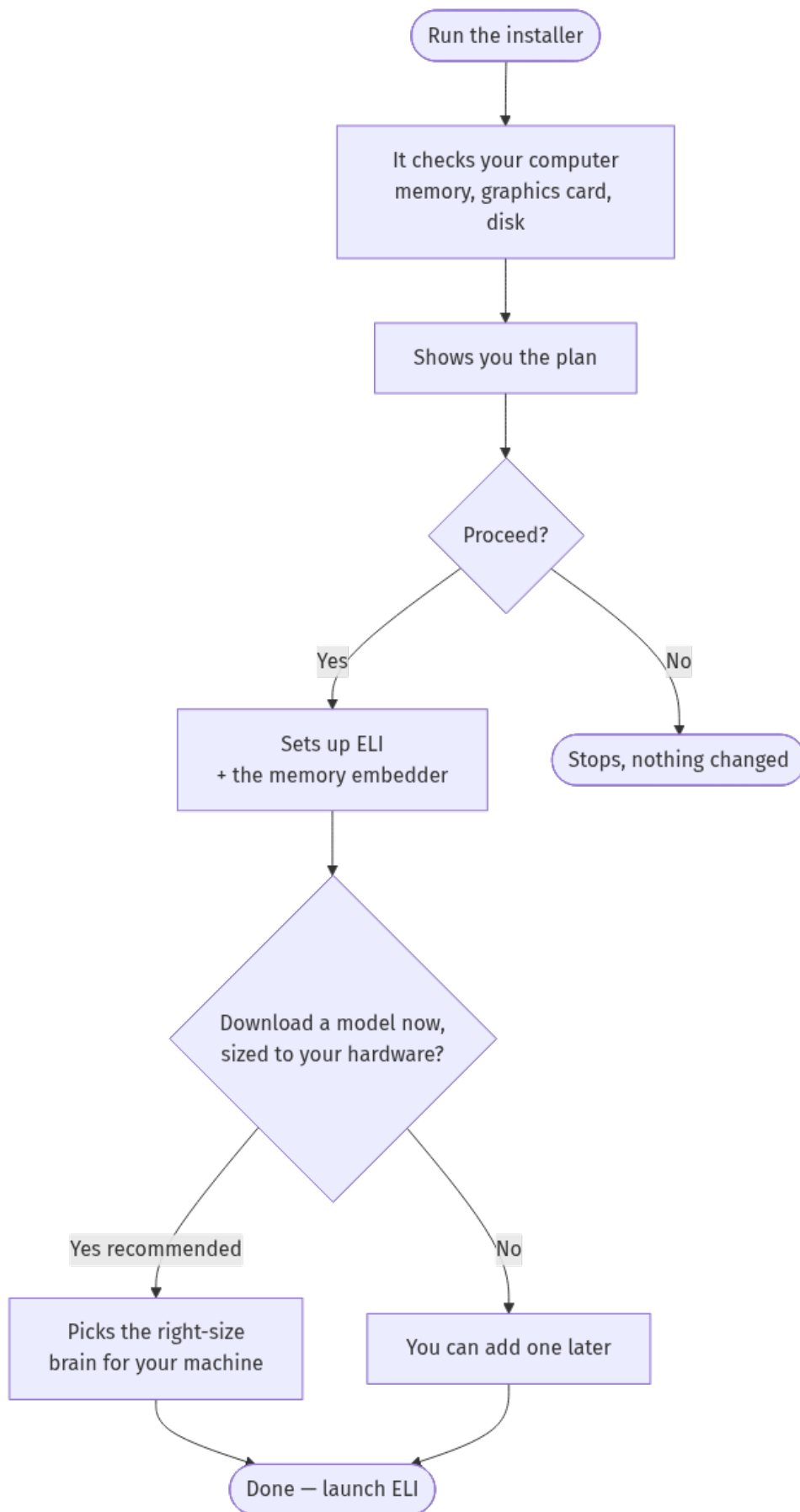
3. Getting started

[A bit technical] *A little technical, but only once.*

Installing

You run a single installer and answer a couple of friendly questions.

- **Windows:** run `install.ps1`
- **Mac & Linux:** run `install.sh`



The installer also fetches a tiny **embedder** model automatically (it powers ELI's memory), so you don't have to think about it. Everything ELI needs ends up in place — no extra steps.

First launch — the quick interview

The very first time you open ELI, it doesn't know you yet, so it asks **three short questions** (your name, what you do, how you like to be spoken to). Ten seconds, and you can **skip it** entirely. From there, ELI builds its understanding of you naturally as you talk.

Opening ELI

- **Desktop app (full experience):** launch **ELI Pro** from your apps menu, or run `scripts/eli_launch.sh` (Mac/Linux) / the desktop shortcut (Windows).
 - **From your phone:** see §13.
-

4. Your first hour with ELI

[Anyone] A gentle on-ramp. Try these in order — each builds confidence.

1. **Say hello.** Type how are you? — get a feel for ELI's voice.
2. **Ask what it can do.** what can you do? lists its capabilities.
3. **Open something.** open the file manager or open firefox.
4. **Play music.** play lo-fi on YouTube, then pause, then skip.
5. **Look at your screen.** what's on my screen?
6. **Tell it a fact.** remember that my dog's name is Rufus. Later: what's my dog's name?
7. **Set a reminder.** set a timer for 5 minutes.
8. **Get a briefing.** what's the news? (this one goes online — ELI will say so).
9. **Check on ELI itself.** what are you running on? shows its model and hardware.
10. **Go hands-free.** Say the wake word **“computer”**, then a request.

By the end you've touched conversation, control, vision, memory, reminders, the web, voice, and introspection — the whole surface.

5. Talking to ELI

There are two ways, freely mixed:

- **Type** in the chat box and press Enter.
- **Speak** — click the microphone, or say the **wake word** (default **“computer”**) then your request. Change the wake word to anything (§11).

You don't need special phrasing. Talk normally:

“open spotify and play some lo-fi” “what's on my screen?” “remember that my sister's name is Anna” “set an alarm for 7am”

Chaining: ask for several things at once — > “close steam and set an alarm for 7am” > “open spotify then play Juicy by Notorious B.I.G.”

If ELI isn't sure what you meant, it asks — it won't guess wildly or pretend. And if a request needs the internet, it tells you before going online.

6. A tour of the window

[Anyone] ELI Pro is organised into **tabs**. You'll mostly live in **Chat**; here's the rest in plain terms:

Tab	What it's for
Chat	Talk to ELI, by voice or text. Home base.
Memory	Browse what ELI remembers about you; add or remove facts.
Habits	Routines ELI has learned; turn on/off, edit times, add your own.
Proactive	Controls for ELI's "act on its own" behaviour (suggestions, summaries, insights).
Tasks	Scheduled & overnight jobs and their status.
Images	Generate images — procedural (fast) or diffusion (photoreal).
Screen	Watch/read your screen, take screenshots, OCR.
Coding / IDE	Write, examine, and fix code; a built-in editor.
Self-Improve	What ELI is proposing to improve in <i>itself</i> — approval-gated.
Report Builder	Generate longer documents/reports with sources (evidence → outline → draft → revise).
ELI's World	ELI's own world-model view.
Labs	Power-user workshop — notebook, Jupyter, calculator, physics, file-chat, projects/workspaces , sim-IDE, orchestration, test review.
Settings	Model (incl. the model-switch dropdown), hardware, voice, and privacy options (§16).

You rarely need the tabs directly — almost everything is reachable by **asking** in Chat: "show my habits", "what do you remember about me", "show background jobs", "make an image of ...", and ELI opens or reports the right thing.

7. What ELI can do — by everyday task

[Anyone] Real things you can say. ELI understands many phrasings — these are examples.

Music & media

- "play lo-fi on YouTube" · "play Juicy by Notorious B.I.G. on Spotify"
- "pause" · "skip" · "next song" · "previous track" · "shuffle" · "repeat this song"
- "what's playing?" · "stop the music" · "skip the ad" (YouTube, when skippable)

Apps & windows

- "open spotify" · "launch firefox" · "open the file manager" · "open github.com"
- "close chrome" · "focus the browser" · "switch to workspace 2"
- "tile windows" · "maximise the window" · "show desktop" · "next window"

Volume, typing, mouse

- "volume up" · "mute" · "set volume to 30"
- "type hello world" · "press enter" · "left click" · "move the mouse up"

Files & notes

- “create a file notes.txt” · “make a folder called projects” · “read notes.txt”
- “list the files in ~/Documents” · “write a note saying buy milk” · “search notes for budget”
- “what’s in my clipboard?” · “summarise report.docx” · “convert report.md to pdf”

Writing, documents & code

- “write a report on renewable energy” · “generate a document about X”
- “write a bash script to monitor the GPU” · “solve this: implement a function that ...”
- “fix the bugs in foo.py” · “examine eli/memory/memory.py for errors” → ELI scans, **offers** fixes, and applies them only if you say “yes, fix it”. · “show the diff”
- The **Report Builder** tab turns “generate a document on X” into a real report: it gathers evidence, plans an outline, drafts each section, then **reviews and revises** it — with document-type quality profiles. It’s generation-first, not a one-shot dump.

Screen, images, PDFs, spreadsheets

- “what’s on my screen?” · “find the submit button on screen” · “take a screenshot”
- “describe cat.jpg” · “read the text in image.jpg” (OCR)
- “summarise report.pdf” · “analyse data.csv” · “watch my screen” / “stop watching”

Information

- “what time is it?” · “what’s the date?” · “weather in Wexford”
- “what’s the news?” / “catch me up” → a **synthesised** briefing, not a raw dump
- “search the web for X” (goes online — ELI tells you) · “gpu status” · “system stats”

Reminders, timers, calendar, scheduling

- “set an alarm for 7am” · “set a timer for 10 minutes”
- “add an event tomorrow at 3pm” · “list my events” · “start a pomodoro”
- “research the best solar inverters overnight” · “build me a script at 2am” → **background jobs**
- “show background jobs” · “check job 5”

Asking ELI about itself (honest introspection)

- “what are you running on?” (model, context, GPU) · “how does your memory work?”
- “what do you know about me?” · “what have you been working on?” · “status”

□ Eye control (gaze)

- “calibrate gaze”, then “enable gaze”. With it on, ELI clicks **where you’re looking** when you say “open” / “left click” / “right click” / “hit enter” — hands-free pointing.
- “gaze status” · “disable gaze”

□ Making images

- “make an image of a red bike by the sea” · “generate a picture of ...”
- Two engines: a fast **procedural** one (always available) and **diffusion** (SSD-1B) for photoreal results — ELI briefly frees the model’s VRAM to run it. The **Images** tab has the controls; results save locally.

Coding & building things

- “solve this: implement a function that ...” → ELI plans it, writes it, then **verifies and repairs** it.
- “generate a project that does X” (a full multi-file scaffold) · “write a bash script to ...”
- “examine eli/memory/memory.py for errors” → a tiered scan; it **offers** fixes and applies them only if you say “yes”. · “show the diff” · “generate tests for your code” (it writes *and* sandbox-verifies them). The **Coding** tab and the built-in **IDE** are home for this.

ELI improving itself

- “improve yourself” → a self-improvement patch cycle — proposed, and **gated by your approval**.
- “show your self-improvement log” · “patch yourself” · “upgrade yourself” (git pull → deps → rebuild indexes) · “run your self-tests”. The **Self-Improve** tab shows what it’s proposing.

□ Plugins & your own agents

- “list plugins” · “enable the X plugin” · “install the X plugin” · “plugin status”
- You can **build your own agent** just by describing it — ELI runs a short dialog, validates it, and registers it live (the create-agent flow).

□ Overnight & scheduled work

- “research the best solar inverters overnight” · “build me a script at 2am” · “every night, regenerate the test report” → durable **background jobs** that survive restarts.
- “show background jobs” · “check job 5”. The **Tasks** tab lists them.

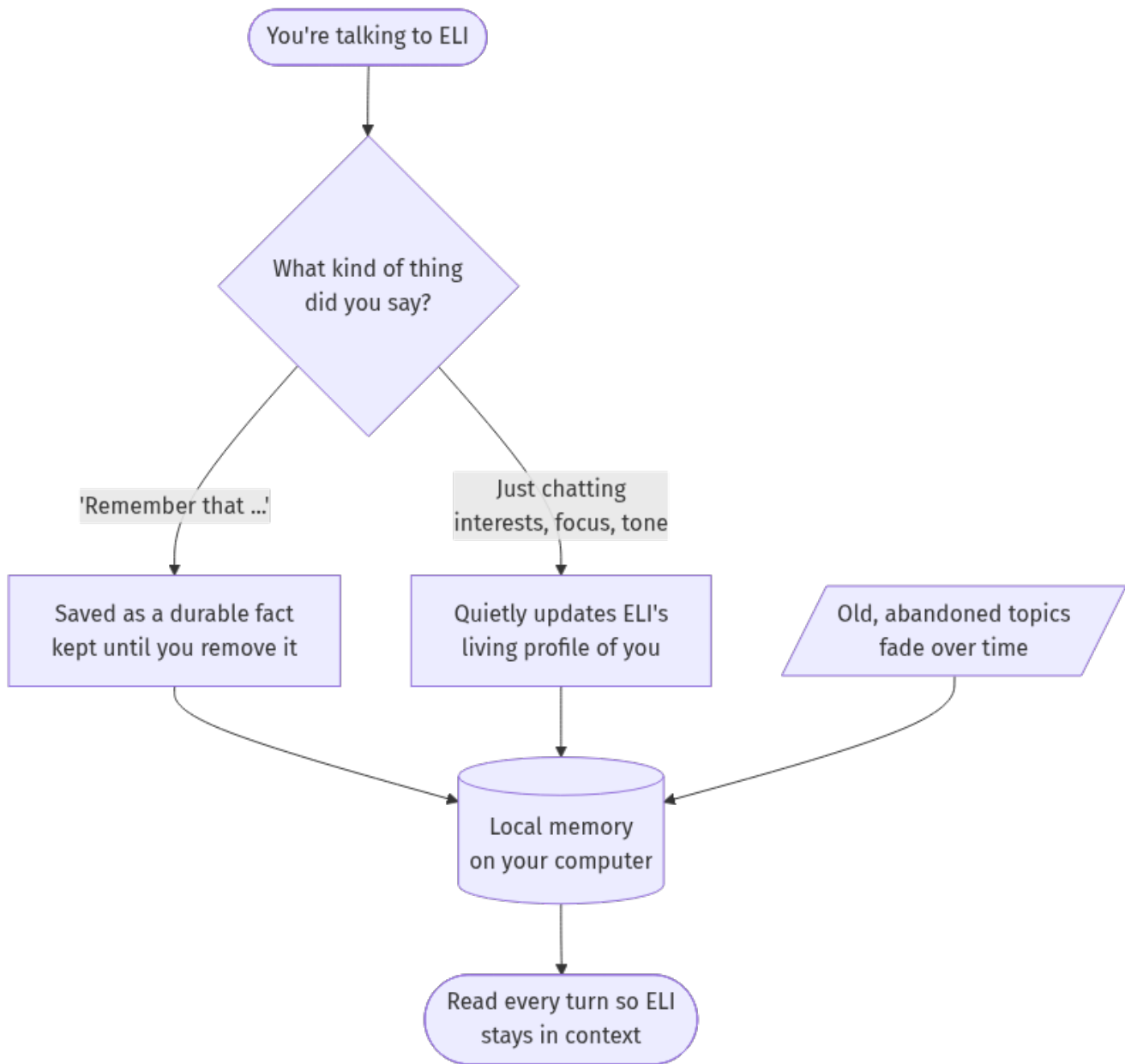
□ Shaping ELI’s voice (persona)

ELI’s personality is **emergent** — it forms from your profile, your history, and the way you talk to it, not from a hand-written script. Most of the time you leave it alone and it just *becomes* itself with you. When you want to steer it, you can, without erasing that: - “lock your persona to a terse senior engineer” · “what persona are you locked to?” · “clear persona lock”. A lock **layers on top** of the emergent voice for the session; clearing it hands the wheel back to the natural persona. - “refresh your persona” reloads it from your latest profile (after ELI’s learned something new about you).

The rule of thumb: **the emergent voice is the default and the truth of who ELI is with you; a lock is a temporary hat it wears on request**. ELI won’t drift into a scripted character on its own, and a lock never overwrites the underlying personality — it sits on top until you clear it.

8. ELI remembers you (memory)

[Anyone] ELI builds a private, evolving understanding of **who you are** and reads it every time you talk, so you never repeat yourself.

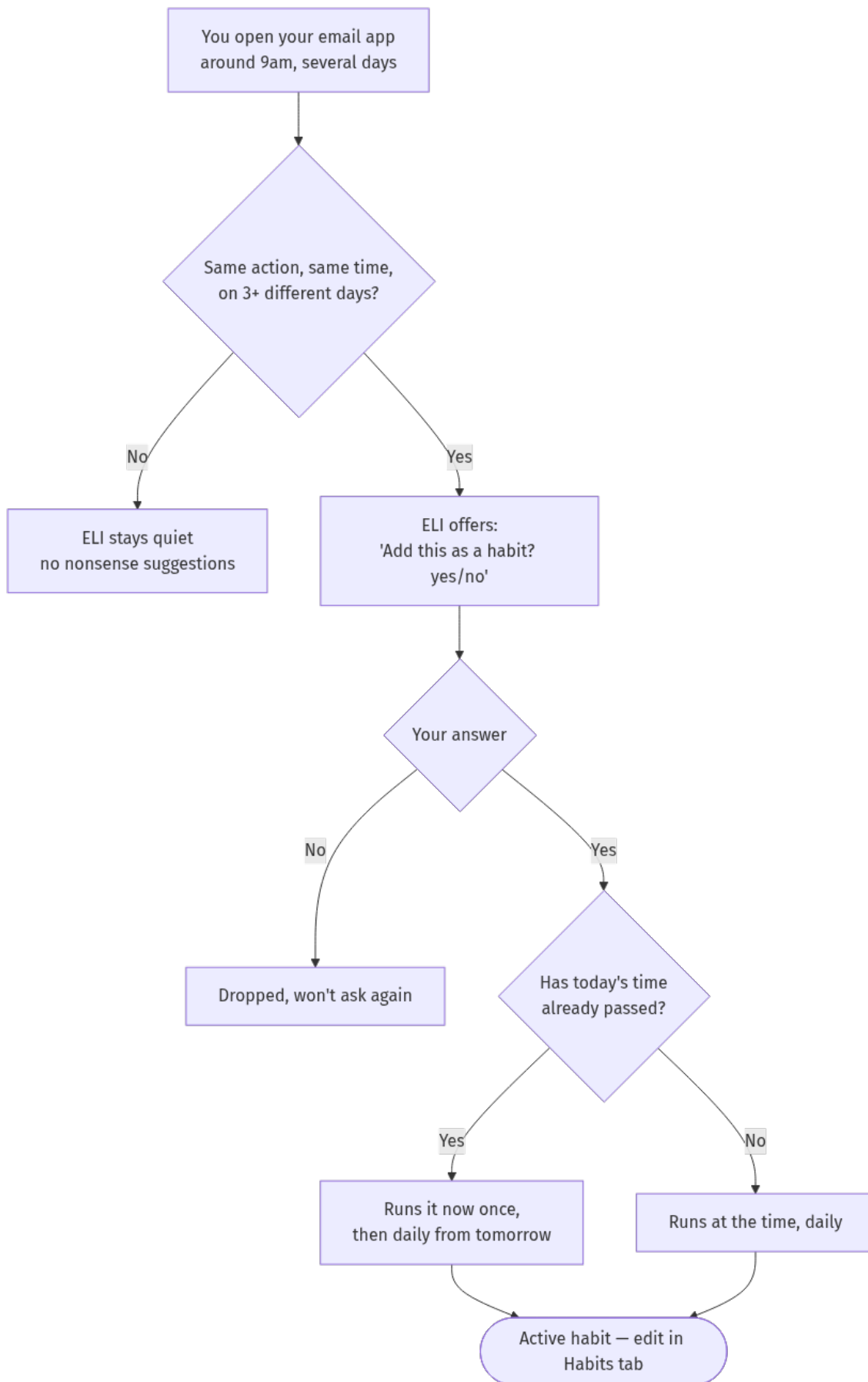


Tell it something to keep: “remember that my sister’s name is Anna” · “my name is Alex” **Ask what it knows:** “what do you remember about my car?” · “what do you know about me?” · “explain everything you know about me and where it’s stored” (a sourced, detailed answer).

Everything is stored **locally** in small private databases. Review and prune it in the **Memory** tab. A brand-new install is a **blank slate** — ELI knows nothing about you until you talk.

9. ELI learns your routines (habits)

[Anyone] If you do the same thing at roughly the same time on **several different days**, ELI notices and **offers** to automate it. It never activates anything without your “yes”.



Manage everything in the **Habits** tab (add, edit times, turn off), or say “show my habits”. The key safeguards: it only proposes routines you’ve **genuinely repeated across days**, and it **always asks first**.

10. ELI helps before you ask (proactive)

[Anyone] With **proactive mode** on, ELI can do helpful things on its own — a **morning briefing**, a gentle suggestion, or a goal it thinks would help. It’s conservative and easy to control:

- “start proactive mode” / “stop proactive mode” · “proactive status”
- “morning report” / “give me my briefing” (news, weather, your agenda)
- “do you have any proposals for me?” / “what’s on your agenda?”

Schedule real unattended work too: > “research X overnight” · “generate tests for your code tonight”
These appear in the **Tasks** tab and survive restarts.

The autonomy tick — what “on its own” actually means

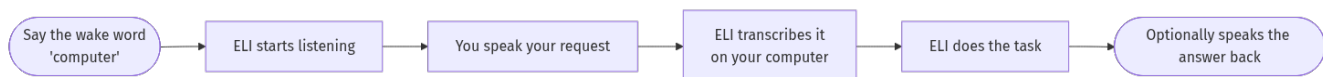
When you allow it, ELI runs a quiet **autonomy tick** inside the proactive daemon — roughly every half hour, and only if it’s switched on (kill switch: `ELI_AUTONOMY_TICK=0`). One tick does three things, all **read-only or proposal-only**:

1. **Code monitor** — it glances over its own code health and flags anything that’s drifted.
2. **Self-model refresh** — it updates its understanding of itself (what it’s running, what it can do) so its introspection stays honest.
3. **Goal autogenesis** — it may propose a goal, or a scheduled task, it reckons would help you.

The guardrail that matters: every autogenerated goal is **proposal-only**. It shows up as a suggestion you accept or dismiss — nothing destructive ever runs unattended, and anything risky still goes through the **approval gate** (Appendix C). Autonomy makes ELI *thoughtful* on its own, not *loose*.

11. Voice, wake word, and dictation

[Anyone] ELI is fully usable hands-free.



- **Wake word:** say “**computer**” then your request. Change it: “change the wake word to athena”. ELI trains its listener on the new word automatically — works even over background music.
 - **Personalise to your voice:** “enroll my wake word” (records you saying it and adapts).
 - **Dictation:** “start dictation” / “stop dictating” — speak and ELI types into the active field.
 - **Transcribe a recording:** “transcribe audio.wav”
 - **ELI speaking back:** “say good morning”
 - **Teach ELI your voice & tone:** “train my voice” — it learns your pitch/energy and even emotion cues, then adapts how it speaks to you.
 - **If voice misbehaves:** “run voice diagnostics”.
-

12. Seeing your screen and images (vision)

[Anyone] ELI can actually **look**:

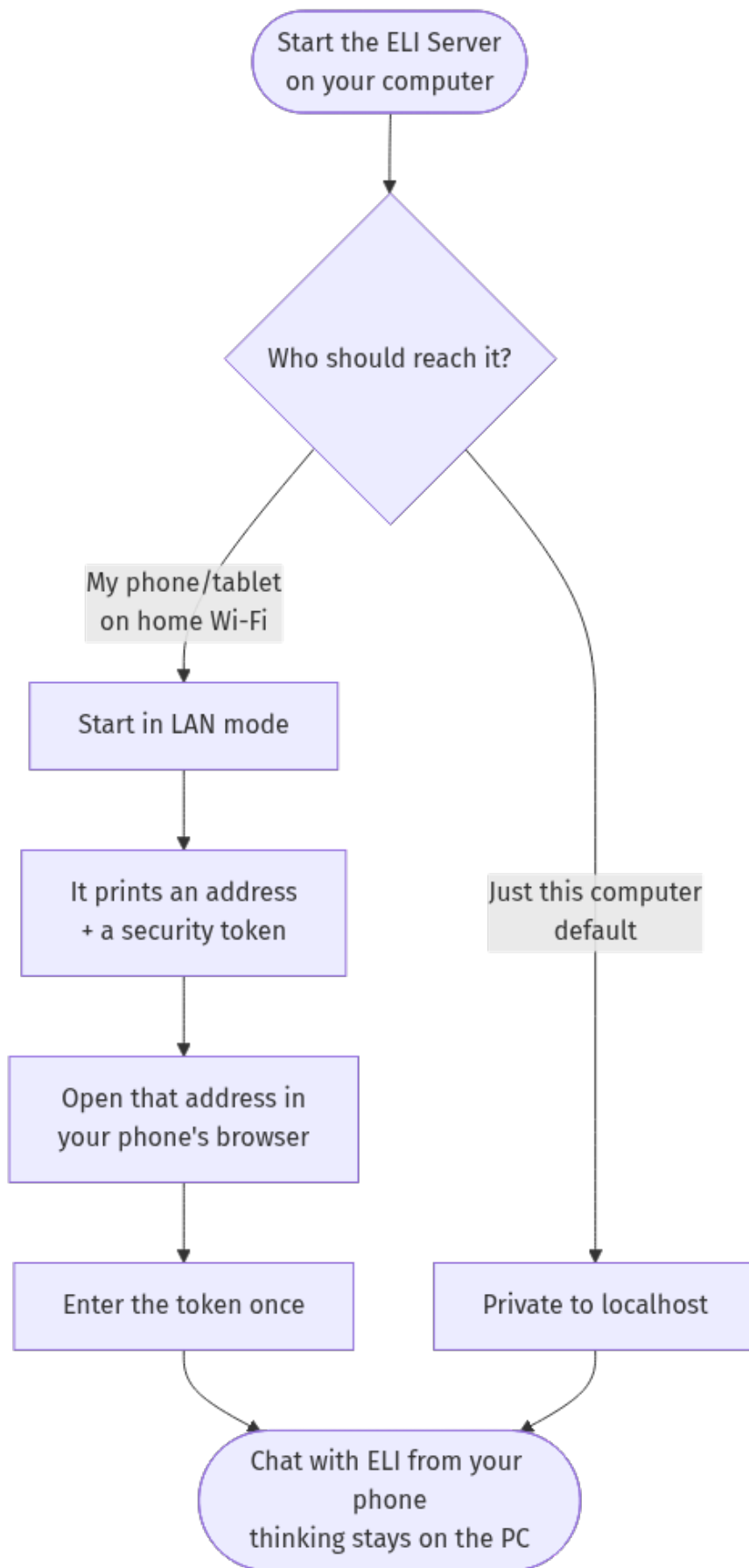
- **Your screen:** “what’s on my screen?” · “what does this error say?” · “find the submit button”
- **Image files:** “describe cat.jpg” · “read the text in this image” (OCR)
- **Documents:** “summarise report.pdf” · “analyse data.csv” · “analyse the pdfs in that folder”
- **Continuous watching:** “watch my screen” (and “stop watching”)

The first time you use vision, ELI may fetch a small vision model (you can pre-download it — §14). With a webcam set up, ELI even supports **eye-tracking control**: “enable gaze control”, “calibrate gaze”, then click where your eyes rest.

13. The ELI server & using it from your phone

[A bit technical] *Slightly technical, but only at setup.*

ELI includes a built-in **server** and **web app**. Start the server on your computer and chat with ELI from your **phone or tablet’s browser** over your home Wi-Fi — while all the actual thinking stays on your computer.



- Start it from the desktop app — **Settings → Web Server** (a “this computer” button and a “phone / Wi-Fi” button that shows the exact link to open). Or run `scripts/eli_serve.sh` (Mac/Linux) / `scripts/eli_serve.ps1` (Windows). Full details: `docs/SERVER_AND_WEB_APP.md`.
- The web app is **installable** — your phone’s “Add to Home Screen” makes it open like an app, and the shell works offline. Light/dark themes; you can recolour the chat.

Starting the server, and what it prints

Start the server and it does two friendly things so you never have to hunt for the address: it **prints a clickable link** (Ctrl/Cmd-click it right in the terminal), and it **auto-opens your local browser** to the dashboard. (Set `ELI_API_NO_BROWSER=1` if you’d rather it didn’t.)

- **This computer only (default):** the dashboard opens at `http://127.0.0.1:8081/`. No token — you’re on the machine, so you’re trusted automatically.
- **Phone / Wi-Fi (LAN mode):** it prints your computer’s LAN address (e.g. `http://192.168.1.20:8081/`) with a **one-time token** baked into the link, plus a **QR code** in the Connect tab. Scan it or open the link and you’re in.

The token, and the rotate button

The phone’s token is **stable** — it’s saved (permissions `0600`, under your config dir) and survives server restarts, so a paired phone stays paired; no re-scanning every reboot. Lost the phone, or just want a clean slate? Hit **rotate**: it mints a fresh token and instantly cuts off anything still using the old one.

Voice from the phone needs HTTPS

Browsers block the microphone on a plain `http://` LAN address, so the phone **mic** needs a secure page. Start with `--https` (or the HTTPS toggle) and ELI runs a *second* server alongside on port **8443** with a self-signed certificate — accept the one-time “not private” warning and the mic works. Plain HTTP stays the primary path (it’s what QR scanners open reliably); HTTPS runs *in addition*, purely for voice.

Switching the model from the dashboard

Settings → Model lists every model installed on your computer, with sizes. Pick one and ELI **hot-swaps** it live — no restart. Only real, installed files show up, so you can’t point it at something that won’t load, and only an **admin** can switch it.

Two servers, one machine

Worth knowing: “the server” is really **two** local services. The **web/API server** (port 8081) serves the dashboard, the phone app, and a small API. The **device server** is a separate MQTT service that talks to your smart-home gear (see the Home tab, below). Both run locally; both are yours. Everything about who’s allowed in — tokens, roles, the loopback trust — is laid out in **Appendix C (Security)**.

If a phone can’t connect

Nine times out of ten it’s the firewall. When it starts in LAN mode ELI prints the **exact** command to open the port for your OS — run that, and make sure the phone is on the **same Wi-Fi**. It’s all scoped to your local network; nothing is exposed to the internet.

What's in the web app

It has a sidebar with several sections: - **Overview** — a live dashboard (clock, GPU/CPU/RAM gauges, status, recent activity). - **Chat** — streamed replies with markdown/code, saved sessions, stop/regenerate, and voice in/out. - **Commands** — a searchable list of everything ELI can do. - **Home** — ELI's own **home-AI** (see below). - **System** — live, measured GPU/CPU/RAM/model telemetry. - **Research** — shared document corpora you can build and query **together**, with citations. - **Audit** — a tamper-evident log of every action. - **Admin** — (admin only) per-user activity, the risk-gate policy, and user management.

Roles (optional)

By default the person on the computer is the operator (full admin). If others share it, you can create accounts with roles — **viewer** (read-only dashboards), **member** (use everything), **admin** (also the Admin console) — in the Admin tab. Each gets a one-time link; the audit trail then attributes actions to the right person.

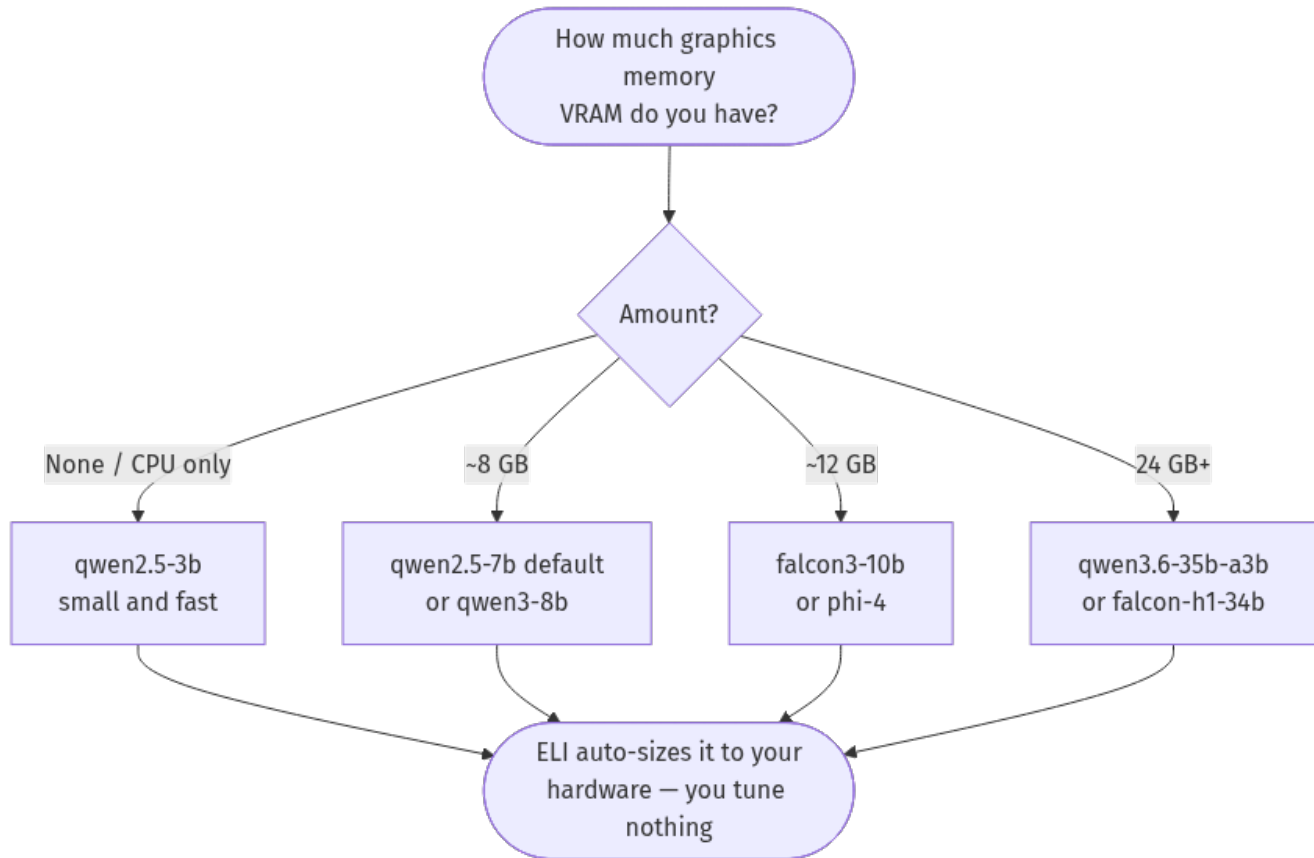
The Home tab — ELI as your home assistant

ELI controls smart devices **directly over MQTT** (ESPHome / Tasmota / Zigbee2MQTT) — no Home Assistant, no cloud: - **Find devices** on your network with one click (mDNS), or add one by its topics. - **Control** lights/switches, group them into **rooms**, and run room-wide on/off — from the app or by voice (“turn on the desk lamp”, “turn off the kitchen”). - **Scenes** — save a set of device states as “Movie mode” and activate it (by voice too). - **Automations** — *when* something happens (a time, **sunrise/sunset**, or a device turning on/off), *do* something (control a device or run a scene), optionally *only if* a condition holds. - ELI **learns** how you use your home and **suggests** automations you can accept with one tap.

14. Models: picking, switching, and training your own

[A bit technical] The **model** is the “brain” ELI thinks with — a file on your computer. Bigger = smarter but needs more graphics memory (VRAM). ELI picks a suitable one at install; change it anytime.

Which model should I get?



Downloading

- **Pick from a menu (choose any number of models):** `python -m eli.core.model_download --choose` — this is what the installer uses; you tick the ones you want (or all / auto / none).
- Let ELI choose one best-fit for your hardware: `python -m eli.core.model_download --auto`
- See the menu without downloading: `python -m eli.core.model_download --list`
- Grab a specific one: `python -m eli.core.model_download qwen2.5-7b`

You're not limited to one — download several and switch between them anytime (§“Switching models”).

Command name	Model	Roughly needs
qwen2.5-3b	small & fast	4 GB graphics / or CPU
qwen2.5-7b (default)	great all-rounder	8 GB graphics
qwen3-8b	newer, big context, good reasoning	8 GB graphics
falcon3-10b	a step up	12 GB graphics
phi-4	strong reasoning for its size	12 GB graphics
qwen3.6-35b-a3b	very capable (mixture-of-experts)	24 GB / or slow on CPU
falcon-h1-34b	largest option	24 GB / or slow on CPU

You can also just **drop any .gguf file** into the `models/` folder and ELI finds it.

Switching models

In **Settings** → **Model** (or “Model menu / Auto Detect”), pick the file. ELI sizes it to your hardware automatically — you don’t tune anything.

[Advanced] Training your own ELI-flavoured model (advanced, optional)

You can fine-tune a model so it speaks in **ELI’s voice** out of the box.

-> **See blueprints/finetuning_guide.md** — a full step-by-step from choosing a base model, through extracting a voice dataset and training, to producing a ready-to-load .gguf. Crucially, ELI’s **personality stays live** (never frozen into the model); training only shapes the *manner* of speech.

15. How hard ELI thinks (reasoning modes)

[Anyone] ELI has five effort levels. Faster modes answer quickly; deeper modes think longer and use more of ELI’s internal helpers for hard problems.

Mode	Feel	Use it for
quick	snappy	greetings, simple commands, quick facts
normal	balanced (default)	everyday questions and tasks
advanced	more thorough	tricky, multi-step asks
research	digs deep	gathering and synthesising a lot
expert	maximum effort	the hardest problems

Ask “what reasoning mode are you in?” or “explain all your reasoning modes”. For most people the default is right — ELI also deepens on its own when a problem clearly needs it.

16. Settings you’ll actually touch

[A bit technical] Most people never need these. The few that matter:

- **Model** — which brain ELI uses (\$14).
- **Voice** — wake word, text-to-speech on/off, microphone.
- **Hardware / startup** — ELI auto-tunes; advanced users can nudge how much graphics memory to hold back, or how much context to use. Leave alone unless you know why.
- **Privacy** — ELI is offline by default; this is where anything network-related lives.

Two graphics cards? ELI can use both — set in the GPU profile settings; single-card just works.

17. Privacy & staying offline

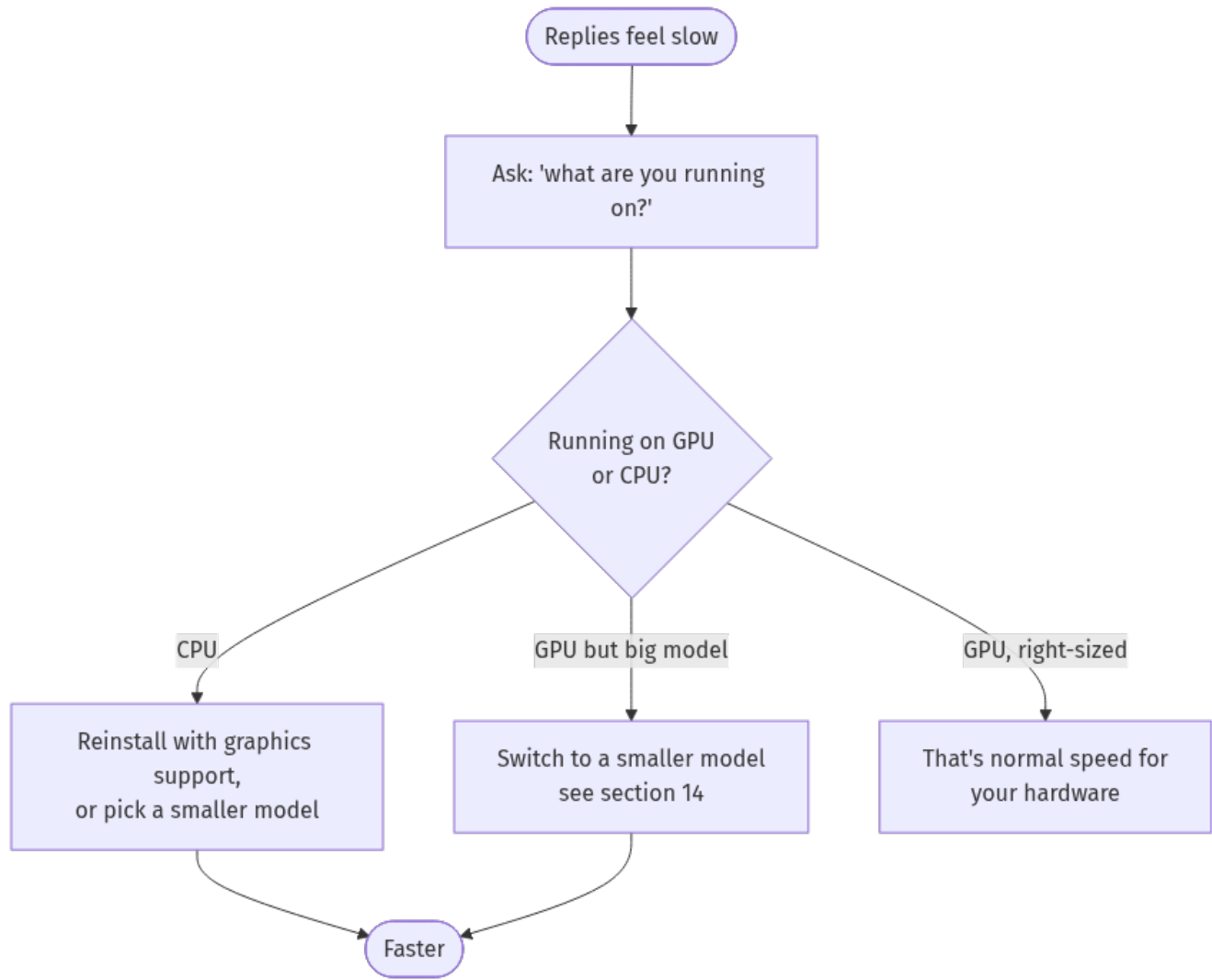
[Anyone]

- **Nothing leaves your computer** unless you ask for something online (web search, news, weather, downloading a model). ELI tells you when it goes online.
 - **Your data is yours** — conversations, memories, files, profile live in local databases. Delete anytime.
 - **No accounts, no telemetry, no subscription.**
 - A fresh install is a **blank slate** — ELI starts knowing nothing about you.
-

18. Troubleshooting

[A bit technical] Most issues map to one of these. When in doubt, **ask ELI** — “run a full runtime audit” or “status” — it reports honestly.

“ELI is replying very slowly”



Everything else

Problem	Try this
Replies look like gibberish	A model whose context is too small. Pick a recommended one (§14); ELI sizes context automatically.
“I couldn’t find a model”	<code>python -m eli.core.model_download --auto</code>
Memory/recall seems empty	Normal on a fresh install. Ensure the embedder downloaded: <code>python -m eli.core.model_download --aux</code>
Voice/wake word not responding	“run voice diagnostics”; check the microphone in Settings → Voice.

Problem	Try this
Won't go online for news/search	By design (offline-first). The network is allowed per-request; check Settings → Privacy.
A habit didn't run	Open Habits — confirm it's enabled and the time is right; or "show my habits".
Something seems broken	"run a full runtime audit" or "status".

19. Quick phrasebook (cheat sheet)

Everyday - "open spotify and play lo-fi" · "pause" · "skip" - "what's on my screen?" · "summarise report.pdf" - "set an alarm for 7am" · "set a timer for 10 minutes" - "what's the news?" · "weather in Dublin" · "what time is it?"

Memory & you - "remember that ..." · "what do you know about me?" · "my name is Alex"

Habits & proactive - "show my habits" · "yes" / "no" (to an offer) · "morning report" · "start proactive mode"

Writing & code - "write a report on X" · "write a script to ..." · "fix the bugs in foo.py"

Voice - wake word "computer" + request · "change the wake word to athena" · "start dictation" · "say good morning" · "train my voice"

Check on ELI - "what are you running on?" · "status" · "what have you been working on?"

Models (in a terminal) - `python -m eli.core.model_download --list` · `python -m eli.core.model_download --auto`

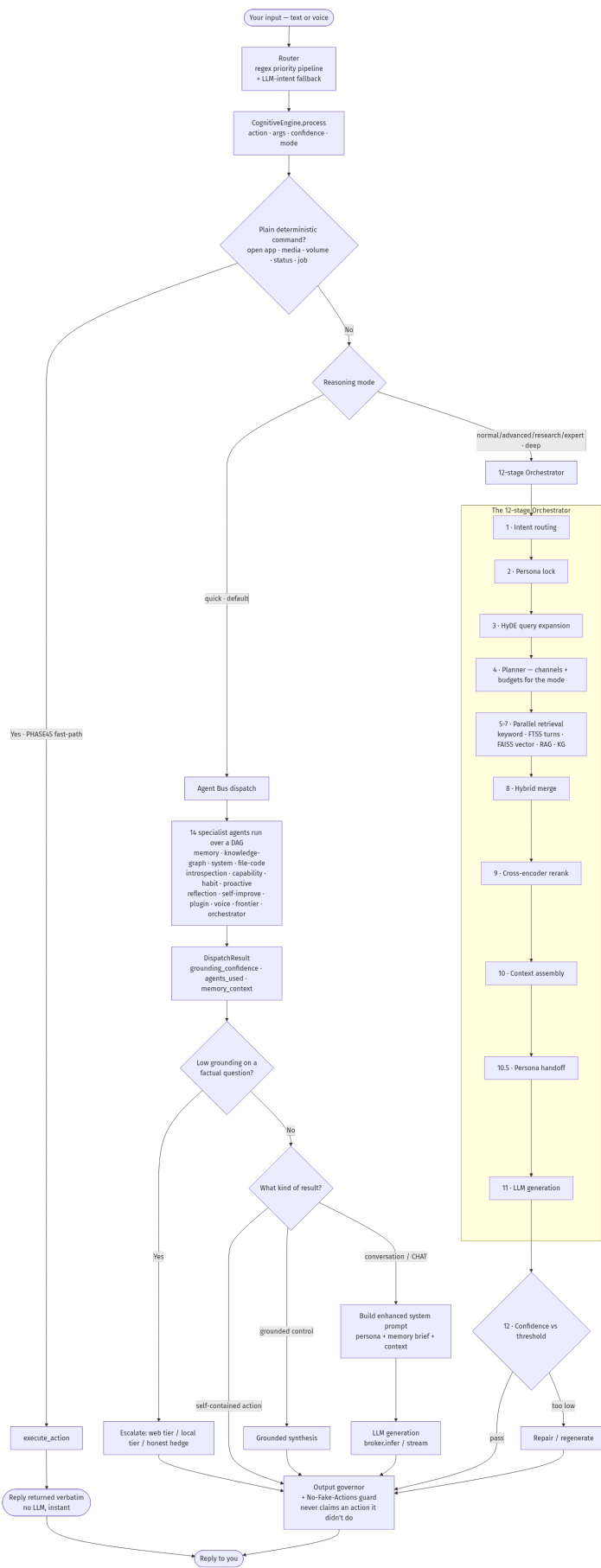
20. Glossary

Term	Plain meaning
Model / GGUF	The "brain" file ELI thinks with. .gguf is just its file format.
VRAM	Memory on your graphics card. More VRAM → you can run a bigger, smarter model.
Embedder	A tiny helper model that powers ELI's memory/recall. Installed automatically.
Context	How much of the conversation ELI can "hold in mind" at once. ELI sets this for you.
Vision model	The model that lets ELI understand screens and images.
Wake word	The phrase that makes ELI start listening (default "computer").
Proactive mode	ELI doing helpful things on its own (briefings, suggestions).
Habit	A routine ELI learned and you approved, run on a schedule.
Fine-tuning	Optional advanced step to teach a model ELI's voice (\$14).
Offline-by-default	ELI never uses the internet unless you ask for something that needs it.

21. Appendix A — Under the hood: how ELI thinks (the full pipeline)

[Advanced] *For the curious / technical. You never need this to use ELI — but here is the complete, verified path from your words to ELI's reply.*

Every request runs through one funnel — `CognitiveEngine.process()` — which takes one of three paths depending on what you asked and the reasoning mode. The diagram is the **real** pipeline (verified against the running code): a deterministic fast-path for plain commands, a **quick** path via the Agent Bus (the default), and a **deep** 12-stage Orchestrator for hard work.



The retrieval stages (5-7) are where memory plugs in — that’s Appendix B. The two guarantees baked into the end of every path: the **Output governor** cleans the reply, and the **No-Fake-Actions guard** ensures ELI never claims it did something it didn’t.

A.1 — The router (before any path is chosen)

Your words first hit the **router**. It runs a **regex-first priority pipeline**: an ordered set of fast, exact checks (web realtime lookups, media controls, file/PDF, memory, OS control, ...), each carefully shaped so it only fires on a real command — e.g. media controls need a whole word in command shape, so “metabolise” never triggers a media “tab”. If nothing matches, an **LLM intent classifier** is the fallback. The router emits {action, args, confidence} and hands off to `CognitiveEngine.process()`. The router also remembers the *last used path* so follow-ups (“do it again”, “the second one”) resolve correctly.

A.2 — The three paths, and when each is taken

- **Fast-path (PHASE45) — no AI involved.** Plain, deterministic actions (open an app, pause music, set volume, report status, check a job) are executed directly and the result is returned **verbatim**. There’s no model call, so these are instant and can’t be “hallucinated”. This is why “pause” never produces a chatty paragraph.
- **Quick path (the default) — the Agent Bus.** Conversational and lightly-grounded requests go to the **Agent Bus**, which runs up to **14 specialist agents** in parallel over a dependency graph, each contributing evidence (memory hits, knowledge-graph facts, system readings, code search, capability info, ...). Their combined result carries a **grounding confidence** — an honest “is this actually backed by anything?” score.
- **Deep path — the 12-stage Orchestrator.** When you ask for depth (advanced/research/expert modes, or a hard question), ELI runs the full pipeline below: more retrieval, reranking, and a confidence check with automatic repair.

A.3 — The Agent Bus, in full

The bus **selects a relevant subset** of agents for your intent (or fans out broadly when unsure), runs them concurrently, and **aggregates** their findings into a `DispatchResult`. Key agents and what they bring:

Agent	Contributes
Memory	recalls facts/turns from SQLite + full-text + vector search
Knowledge-Graph System	entities and how they relate (who/what/where) runs executor actions (web search, audits, status)
File-Code	searches the actual codebase for grounded answers about ELI itself
Introspection Capability	live runtime/self state what ELI can do (the capability manifest)
Habit / Proactive / Reflection / Self-Improve Plugin / Voice / Frontier / Orchestrator	routines, signals, insights, fixes plugin dispatch, speech state, deeper reasoning hooks, planning

After the bus returns, a **grounding-escalation hook** checks: *is this a factual question that came back weakly supported?* If so, ELI escalates — a web tier (if you allow online), a local tier, or an **honest hedge** (“I’m not certain, but...”) rather than a confident guess. Then the result is finalised: a self-contained action returns directly, a grounded control action gets a short grounded synthesis, and a conversation goes to generation with the full persona + memory context.

A.4 — The 12 stages, one by one (the deep path)

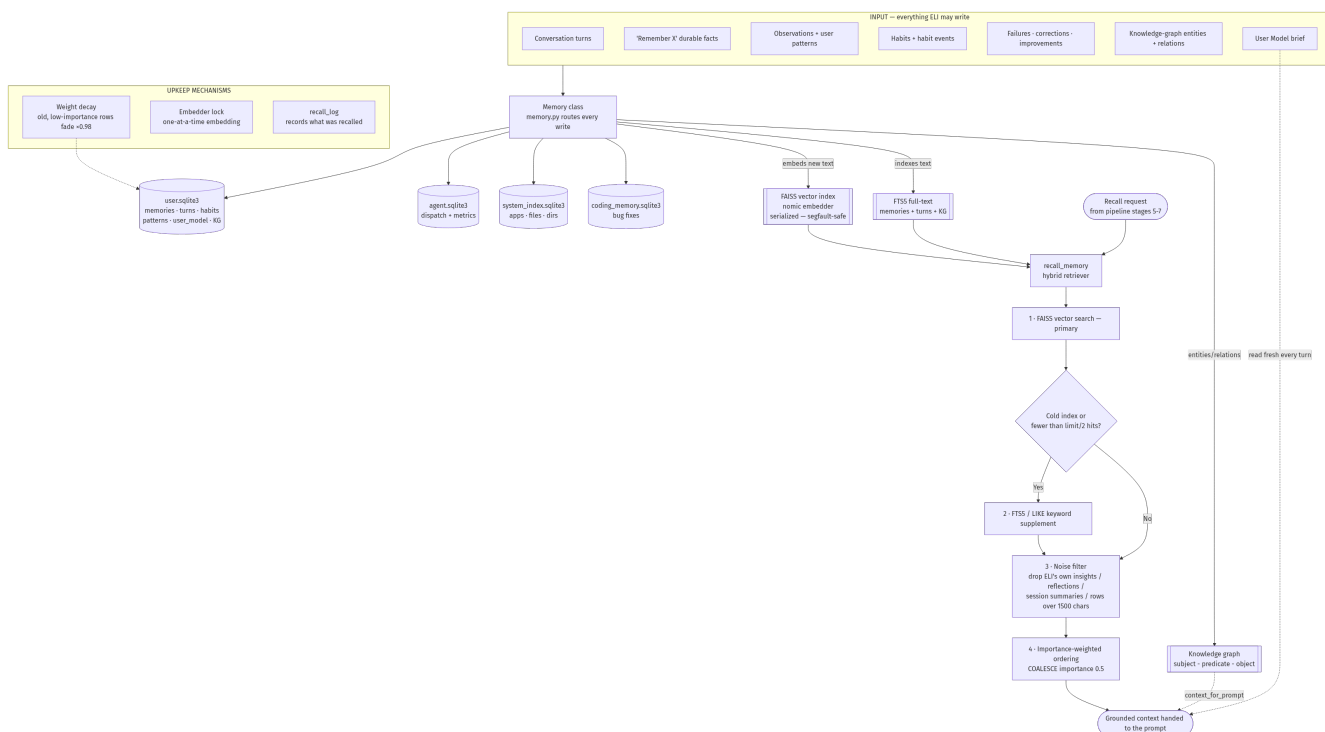
1. **Intent routing** — settle exactly what you're asking for.
2. **Persona lock** — fix ELI's identity/voice for this turn so it stays consistent.
3. **HyDE query expansion** — ELI drafts a *hypothetical answer* and searches with **that** too, which dramatically improves what memory retrieves (you find better matches by searching with an answer-shaped query, not just the bare question). Skipped for very short queries.
4. **Planner** — decide *which* retrieval channels to use and how big the budgets are, tuned to the reasoning mode (fast / balanced / deep). 5–7. **Parallel retrieval** — several searches run at once: keyword, **full-text (FTS5)** over your conversation history, **FAISS vector** (semantic) search, optional RAG, and the **knowledge graph**. (This is the seam where Appendix B's memory system feeds in.)
5. **Hybrid merge** — combine the hits from every channel into one candidate set.
6. **Cross-encoder rerank** — a smarter model re-scores the merged candidates for true relevance to your question, and the best rise to the top. (Skipped for non-conversational asks.)
7. **Context assembly** — build the grounded context block that will inform the answer. 10.5. **Persona handoff** — wrap that context with ELI's live persona brief for generation.
8. **LLM generation** — the model writes the reply (streaming or one-shot). For the deepest modes this is a two-part private-reasoning-then-condense step.
9. **Confidence check** — score the answer against a threshold; if it's too low, ELI **re-pairs/regenerates** rather than shipping a weak answer.

A.5 — The two end-of-path guarantees

Every path, no matter which, ends at: - **Output governor** — normalises and cleans the reply (strips artefacts, enforces formatting). - **No-Fake-Actions guard** — if ELI's reply *claims* it did something (played a song, fixed a file), the system verifies it actually happened; if not, the claim is replaced with the honest truth. This is enforced in code, not left to the model's goodwill.

22. Appendix B — Under the hood: how ELI remembers (the full memory system)

[Advanced] *For the curious / technical. ELI's memory is a genuine hybrid — relational + full-text + vector + graph — all local. This is the complete input → storage → recall path (verified against the code).*



Why it's trustworthy: the **noise filter** stops ELI's own past replies and reflections from resurfacing as if they were *your* facts; the **embedder lock** keeps the vector engine from crashing under concurrent use; and **weight decay** is the gentle "forgetting" that lets stale topics fade while active ones stay fresh. Nothing here leaves your computer.

B.1 — What ELI writes (the inputs)

Everything that flows into memory passes through one **Memory** module, which decides where each item belongs: - **Conversation turns** — what you said and what ELI replied, with timestamps. - **Durable facts** — anything you explicitly ask it to "remember". - **Observations & user patterns** — quiet signals about your interests, focus, and tone that build the living profile. - **Habits & habit events** — your repeated actions and the routines learned from them. - **Failures, corrections, improvements** — ELI's own mistakes and fixes, kept for self-learning (these are deliberately **excluded** from normal recall — see the noise filter). - **Knowledge-graph entities & relations** — named things and how they connect. - **User Model brief** — a compact, pre-rendered summary of who you are, refreshed over time.

B.2 — Where it's stored (four local databases + three indexes)

Everything is plain local files on your machine — no server, no cloud: - **user.sqlite3** — the main store: memories, conversation turns, habits, patterns, the User Model, and the knowledge graph. - **agent.sqlite3** — ELI's internal agent activity and performance metrics. - **system_index.sqlite3** — an index of your apps, executables, files, and folders (so "open the file manager" is instant). - **coding_memory.sqlite3** — remembered code bug-fixes.

On top of the relational data sit three search indexes: - **FAISS vector index** — every stored text is turned into an "embedding" (a list of numbers capturing meaning) by a small local **nomic embedder**, so ELI can find things by *meaning*, not just exact words. Embedding is done **one-at-a-time** behind a lock (the embedder isn't thread-safe; serialising it prevents crashes). - **FTS5 full-text index** — fast exact/keyword search over memories, conversation turns, and the knowledge graph. - **Knowledge graph** — a subject → predicate → object web (e.g. *you* → *own* → *a Tesla*) that ELI can walk to answer relational questions.

B.3 — How recall works (step by step)

When the cognition pipeline (stages 5–7) needs memory, it calls **recall_memory**, a genuine hybrid retriever that runs in this order: 1. **Vector search first (FAISS)**. The semantic index is the primary path — it finds things that *mean* the same even if worded differently. 2. **Keyword supplement (FTS5/LIKE) — only if needed**. If the vector index is cold/empty or returns fewer than half the requested results (e.g. a very short query), a keyword search tops it up. (When another part of the system already did the vector search, this step is skipped to avoid double-searching.) 3. **Noise filter**. Results are scrubbed of ELI’s own output — assistant insights, reflections, session summaries, the “orchestrator” source, and any over-long blobs (>1500 chars). This is the safeguard that stops ELI’s past musings from masquerading as *your* facts. 4. **Importance-weighted ordering**. What survives is ranked by an importance score so the most significant memories surface first.

Alongside this, the **knowledge graph** contributes a lightweight relational context, and your **User Model brief** is read **fresh every single turn** — that’s why ELI stays in context about who you are without you repeating yourself.

B.4 — Upkeep (the mechanisms that keep memory healthy)

- **Weight decay** — old, low-importance memories are gently faded (multiplied down over time), so abandoned topics naturally recede while active ones stay prominent. This is ELI’s “forgetting”.
- **Embedder serialization** — the single lock around the embedder that keeps the vector engine stable under concurrent access.
- **Recall log** — a record of what was retrieved, used for tuning and for honestly explaining *where* a piece of knowledge came from (“how do you know that?”).

B.5 — Privacy, restated

Every database and index in this appendix is a local file under your user profile. There is no sync, no upload, no account. Delete the files (or use the **Memory** tab) and that knowledge is gone. A fresh install starts with the full structure but **zero** personal rows — a true blank slate.

23. Appendix C — Under the hood: how ELI stays safe and yours (security)

[Advanced] *For the curious / technical. ELI is meant to be safe by construction, not by good intentions — every guard below is enforced in code, not left to a setting you might forget to flip. All of this is verified against the running codebase.*

C.1 — Offline by default (the socket failsafe)

ELI doesn’t just *prefer* to stay offline — it’s physically stopped from reaching the internet unless you asked for something that needs it. At startup, netguard installs a **process-wide socket guard**: it patches the low-level `connect()` call so that *any* outbound connection to a non-loopback address raises an `OfflineError` while ELI is offline. Loopback and local sockets (the phone web server, the local model) always work. So even if some stray bit of code *tried* to phone home, the operating-system call itself is blocked — belt and braces. When you do turn networking on for a task, that flip is written to the audit ledger; “internet on” is never silent.

Ask for something online while offline and you don’t get a crash or an invented answer — you get an honest refusal: “*I can’t fetch that right now, I’m offline.*”

C.2 — The fail-closed command gate

ELI can run shell commands and open apps, so that path is gated hard. Before any command runs it’s checked against an **allowlist** by the `SecurityManager`. The part that matters: if the security layer

can't load for any reason, the check **fails closed** — it returns *denied*, not *allowed*. A broken guard blocks the action; it never quietly waves one through. Sandboxed runs are limited to a fixed set of allowed executables.

C.3 — The approval gate (nothing risky happens behind your back)

Anything ELI proposes on its own — an autonomous action, a self-improvement patch, a scheduled job — goes through the **approval engine** before it can execute. Each proposal carries an *action class*, and the engine decides: auto-approve the harmless ones, or hold the riskier ones until **you** confirm. Who is even allowed to propose what is also gated — a background daemon can't quietly emit a dangerous action. There's a **Full Control** mode that lifts the confirmation barrier if you want ELI to just act, but that's your explicit choice, and it's off by default.

C.4 — The phone: who's allowed in

The web server (§13) is locked down. On the same machine (loopback) you're trusted automatically. From the network, every request must carry a **bearer token** — and that token is never stored in plain text, only its SHA-256 hash. Define one or more users and **RBAC** switches on: each token maps to a role (admin / member), and admin-only actions — changing settings, switching the model — reject anyone who isn't admin. With no users defined it's single-operator mode: just you. The **rotate** button issues a fresh token and kills the old one, so a paired phone can be cut off in one tap.

C.5 — The audit ledger

Security-relevant events — settings changes, the network being turned on or off, research collaborations — are written to a **tamper-evident audit ledger**: a local, append-only trail so there's always an honest record of what changed and when.

C.6 — Privacy, restated

Every guard, token, ledger, and database above is a local file under your own user profile. No account, no sync, no telemetry. ELI's default posture is *closed* — offline, gated, and yours. You open doors deliberately, one at a time.

That's the whole assistant — inside and out. You don't need any of the last three appendices to use ELI; just talk to it in plain English, and ask "what can you do?" whenever you're curious. Everything here happens on your computer, for you, privately.